Supplement: Independent Mechanism Validation

17 September 2026

S1 Scope, aggregation, and audit

This supplement accompanies the English mechanism working manuscript. All printed numbers derive from the completed independent validation; development data are not pooled into its tests. The full design comprises 24 BBOB functions, dimensions 10/20/40, instances 11/12/13, three repeats, six methods, and budget 10D: 3,888 runs, 907,200 paid algorithm calls. Independent diagnostics comprise 7,776 states and 1,990,656 additional labels, never supplied to optimizers. Failed runs and exact duplicate calls are zero.

Endpoint and contrast tables use the frozen hierarchy: three-repeat median within instance, three-instance average within function, then equal function weights for suite summaries. Contrasts are constructed at their prespecified component level. Raw-error IQR tables instead describe all nine paid runs for a function and method; they are not the estimators used for primary hypothesis tests.

Only the nine prespecified comparisons form the primary Holm family. Tables here provide complete function-level inputs and descriptive secondary outputs; they add no tests. The observed training truncation count is 185,328 fits. Source, pre-information, original residual, candidate-selection, and diagnostic-index audits passed. Real-arithmetic ordering equivalence is separate from arbitrary floating-point tie behavior.

The original optimization-improvement gate remains failed. New instances support mechanism and failure analysis. Neither this supplement nor the manuscript claims a pointwise coverage theorem, FDP/FDR control, evaluation savings, or application acceleration. Live GP-EI/TuRBO batches and unexecuted exclusive-cost measurements are absent from these tables.

All raw runs, predicted-before-label journals, candidate pools, hidden diagnostic labels, four checkpoint state tables, dependency locks, seeds, and failure logs are available in the project. Seed = 100000D + 1000f + 10i + r, with r=0,1,2. Initialization, candidates, and management use distinct random streams. Sharing this seed alone is not a claim of identical downstream histories.

S2.10 Full function endpoint scores, D=10

Lower is better. Scores are log10(max(final error / common initial error, 1e-12)), with the frozen within-instance median and across-instance average. Random means the paid random candidate policy, not independent-domain Sobol.

Function	Greedy	Fixed .1	Fixed .3	Random	PEM-T	PEM-ST
1	-1.9834	-1.7330	-1.5064	-0.1841	-1.5907	-1.5998
2	-0.5809	-0.7906	-0.7593	-0.0856	-0.5229	-0.5481
3	-0.3652	-0.3124	-0.2679	-0.0609	-0.2810	-0.2941
4	-0.1263	-0.0789	-0.1170	-0.0359	-0.0504	-0.0143
5	-0.8662	-0.8101	-0.7117	-0.0652	-0.7618	-0.7374
6	-3.1709	-3.1172	-3.0736	-0.3935	-2.8907	-3.0354
7	-1.0807	-1.0923	-1.0271	-0.2608	-0.9152	-0.9152
8	-1.7713	-1.8968	-1.6115	-0.3213	-1.4271	-1.5771
9	-1.3192	-1.2243	-1.2079	-0.0788	-1.0173	-1.0684
10	-0.9057	-0.7745	-0.7705	-0.2539	-0.7886	-0.7921
11	-0.8648	-1.2172	-0.6966	-0.7170	-0.6618	-0.7868
12	-0.5551	-0.4166	-0.3526	-0.0371	-0.1603	-0.1835
13	-0.8567	-0.6796	-0.6189	-0.0824	-0.5792	-0.6537
14	-0.7637	-0.7258	-0.5251	-0.1895	-0.4739	-0.4739
15	-0.5217	-0.5104	-0.3960	-0.1514	-0.3795	-0.3632
16	-0.0299	-0.0520	-0.0436	-0.0639	-0.0216	-0.0108
17	-0.3329	-0.3021	-0.2586	-0.1783	-0.2201	-0.2534
18	-0.4174	-0.2357	-0.2848	-0.0877	-0.2106	-0.2106
19	-0.3659	-0.3849	-0.3056	-0.1483	-0.2792	-0.2826
20	-3.4967	-3.3994	-3.1207	-0.4768	-3.3170	-3.4424
21	-0.8677	-0.6352	-0.5508	-0.0107	-0.1617	-0.1242
22	-0.4136	-0.4118	-0.2858	-0.0027	-0.1287	-0.2206
23	0.0000	-0.0887	0.0000	-0.0102	-0.0450	-0.0450
24	-0.3083	-0.2868	-0.2970	-0.1427	-0.2901	-0.2840

S2.20 Full function endpoint scores, D=20

Lower is better. Scores are log10(max(final error / common initial error, 1e-12)), with the frozen within-instance median and across-instance average. Random means the paid random candidate policy, not independent-domain Sobol.

Function	Greedy	Fixed .1	Fixed .3	Random	PEM-T	PEM-ST
1	-1.6820	-1.6604	-1.4000	-0.2050	-1.3338	-1.4436
2	-1.0234	-0.9404	-0.9793	-0.3234	-0.8794	-0.9600
3	-0.4893	-0.4672	-0.4691	-0.1596	-0.4593	-0.4985
4	-0.2492	-0.2330	-0.2305	-0.1605	-0.2004	-0.1837
5	-0.9105	-0.8274	-0.6863	-0.1245	-0.7928	-0.7450
6	-3.3391	-3.4096	-2.7282	-0.3286	-3.1255	-3.1124
7	-0.8569	-0.9348	-0.8613	-0.2253	-0.8318	-0.8752
8	-1.6783	-1.6372	-1.5885	-0.2687	-1.6901	-1.7226
9	-1.3875	-1.3276	-1.2672	-0.4295	-1.2298	-1.3932
10	-0.9261	-0.8323	-0.5536	-0.1740	-0.5420	-0.6041
11	-1.4417	-1.5274	-1.3294	-1.0957	-1.1293	-1.1492
12	-0.3436	-0.3167	-0.2405	-0.3195	-0.3442	-0.3442
13	-0.7274	-0.6725	-0.6150	-0.0814	-0.7002	-0.6307
14	-0.6917	-0.6200	-0.5653	-0.1573	-0.6970	-0.6865
15	-0.5765	-0.5012	-0.5491	-0.2020	-0.4970	-0.5595
16	-0.0482	-0.0863	-0.0652	-0.0889	-0.0937	-0.1138
17	-0.5015	-0.4647	-0.3443	-0.1048	-0.3243	-0.3304
18	-0.4201	-0.3763	-0.3129	-0.1285	-0.3073	-0.3259
19	-0.4221	-0.4355	-0.4064	-0.1518	-0.3917	-0.4100
20	-2.2002	-2.2540	-1.8271	-0.3838	-2.1324	-2.0332
21	-0.7052	-0.6441	-0.7432	-0.0358	-0.3390	-0.2657
22	-0.6147	-0.4840	-0.4384	-0.0143	-0.2533	-0.2130
23	-0.0822	-0.0497	-0.0068	-0.0486	-0.0439	-0.0132
24	-0.3157	-0.3236	-0.2968	-0.1235	-0.2926	-0.2909

S2.40 Full function endpoint scores, D=40

Lower is better. Scores are log10(max(final error / common initial error, 1e-12)), with the frozen within-instance median and across-instance average. Random means the paid random candidate policy, not independent-domain Sobol.

Function	Greedy	Fixed .1	Fixed .3	Random	PEM-T	PEM-ST
1	-1.2112	-1.2326	-1.0894	-0.2584	-1.1720	-1.1741
2	-1.2054	-1.0937	-0.9946	-0.4467	-0.9065	-0.7905
3	-0.5155	-0.5833	-0.4918	-0.1928	-0.5266	-0.5230
4	-0.2900	-0.2612	-0.2655	-0.1742	-0.2393	-0.2376
5	-0.7832	-0.7302	-0.6422	-0.1613	-0.7388	-0.6742
6	-2.6753	-2.8177	-2.4509	-0.3562	-2.3885	-2.2635
7	-0.9055	-0.9318	-0.8804	-0.1851	-0.8669	-0.9004
8	-1.7307	-1.7844	-1.6654	-0.5741	-1.6092	-1.6275
9	-1.2941	-1.2041	-1.1497	-0.5013	-1.1577	-1.0611
10	-1.2909	-1.1676	-1.0811	-0.2766	-1.1969	-1.2051
11	-0.7728	-0.7548	-0.7077	-0.8592	-0.6661	-0.4877
12	-0.5093	-0.4568	-0.3637	-0.3497	-0.3492	-0.3565
13	-0.5333	-0.5347	-0.5095	-0.1336	-0.5020	-0.5148
14	-0.7277	-0.7359	-0.6712	-0.2766	-0.6802	-0.6559
15	-0.4987	-0.5070	-0.4711	-0.2374	-0.4218	-0.4495
16	-0.0602	-0.0521	-0.0575	-0.0487	-0.0365	-0.0773
17	-0.4202	-0.4425	-0.4090	-0.1568	-0.3537	-0.3751
18	-0.5101	-0.4587	-0.4364	-0.2007	-0.4808	-0.4443
19	-0.4884	-0.4515	-0.4362	-0.2502	-0.4420	-0.4647
20	-1.9465	-1.7348	-1.3354	-0.3918	-1.8456	-1.7511
21	-0.5208	-0.3612	-0.3872	-0.0293	-0.3693	-0.3633
22	-0.3394	-0.3528	-0.2130	-0.0249	-0.2416	-0.1843
23	-0.0553	-0.0875	-0.0420	-0.0349	-0.0695	-0.0559
24	-0.2943	-0.3057	-0.2818	-0.1737	-0.2483	-0.2605

S3.10 Full primary contrast inputs, D=10

The three columns have different units. Positive endpoint difference means worse PEM-T; positive gain difference favors feedback. History interaction is a difference of selected-minus-pool failure gaps between fixed .3 and greedy histories and has no inherent favorable sign. It is not a same-pool causal effect.

Function	History interaction	Endpoint T - .1	Gain feedback - mean
1	0.1553	0.1423	-0.0682
2	0.2360	0.2677	0.0000
3	0.1387	0.0314	-0.0104
4	0.1348	0.0285	0.0000
5	0.0889	0.0483	-0.0471
6	0.1833	0.2266	-0.0123
7	0.1742	0.1770	-0.0192
8	0.0944	0.4696	-0.0159
9	0.1911	0.2070	-0.0039
10	0.0586	-0.0141	1.138e-21
11	0.0309	0.5554	0.0000
12	0.0973	0.2564	-0.0002
13	0.0404	0.1005	-0.0540
14	0.0430	0.2519	-0.0067
15	0.0615	0.1309	-0.0207
16	-0.0999	0.0304	0.0000
17	0.1449	0.0820	-1.210e-05
18	0.1891	0.0251	-0.0061
19	0.1432	0.1057	-0.0086
20	0.0150	0.0824	-0.0190
21	-0.0042	0.4735	-0.0138
22	0.0804	0.2831	-0.0592
23	-0.0931	0.0437	0.0000
24	-0.0609	-0.0033	-0.0433

S3.20 Full primary contrast inputs, D=20

The three columns have different units. Positive endpoint difference means worse PEM-T; positive gain difference favors feedback. History interaction is a difference of selected-minus-pool failure gaps between fixed .3 and greedy histories and has no inherent favorable sign. It is not a same-pool causal effect.

Function	History interaction	Endpoint T - .1	Gain feedback - mean
1	0.3343	0.3266	-0.0261
2	0.2367	0.0611	-0.0001
3	0.1042	0.0079	-0.0197
4	0.1204	0.0325	-0.0003
5	0.2581	0.0346	-0.0966
6	0.2796	0.2841	-0.0075
7	0.0749	0.1030	-0.0011
8	0.1976	-0.0529	-0.0393
9	0.1540	0.0978	-0.0025
10	0.1257	0.2903	-0.0001
11	0.0671	0.3980	0.0000
12	0.1185	-0.0275	-0.0001
13	0.2008	-0.0276	-0.0522
14	0.1862	-0.0770	-0.0028
15	0.2533	0.0042	-0.0237
16	-0.2565	-0.0074	0.0000
17	0.0612	0.1404	-0.0046
18	0.0622	0.0690	-2.086e-05
19	-0.0306	0.0438	-0.0022
20	0.2018	0.1216	-0.0250
21	0.0824	0.3052	-0.0327
22	0.0068	0.2308	-0.0041
23	-0.0029	0.0058	0.0000
24	0.1055	0.0310	-0.0389

S3.40 Full primary contrast inputs, D=40

The three columns have different units. Positive endpoint difference means worse PEM-T; positive gain difference favors feedback. History interaction is a difference of selected-minus-pool failure gaps between fixed .3 and greedy histories and has no inherent favorable sign. It is not a same-pool causal effect.

Function	History interaction	Endpoint T - .1	Gain feedback - mean
1	0.3343	0.0606	-0.1179
2	0.2135	0.1873	-0.0003
3	0.1784	0.0567	-0.0233
4	0.1641	0.0220	-0.0114
5	0.3662	-0.0086	-0.1300
6	0.3424	0.4292	-0.0245
7	0.2699	0.0649	-0.0050
8	0.2344	0.1752	-0.0547
9	0.2292	0.0464	-0.0051
10	0.1602	-0.0293	-0.0038
11	0.0671	0.0887	0.0000
12	0.1055	0.1076	1.481e-20
13	0.4954	0.0327	-0.0553
14	0.1465	0.0557	-0.0105
15	-0.0202	0.0852	-0.0031
16	-0.0876	0.0156	0.0000
17	-0.0944	0.0887	-0.0014
18	-0.0352	-0.0222	-0.0030
19	-0.0501	0.0095	-0.0028
20	0.2751	-0.1108	-0.0132
21	0.2386	-0.0081	-0.0375
22	-0.0635	0.1112	-0.3186
23	-0.0023	0.0180	0.0000
24	0.0107	0.0573	-0.0052

S4.10 Raw paid-run error dispersion, D=10

Absolute final error above the known optimum; all nine runs per function and method. This raw dispersion uses a different aggregation from the primary contrasts. Complete six-method IQR values are retained in internal_raw_error_iqr.csv. Scientific notation avoids rounding nonzero errors to zero.

Function	Fixed .1 median [Q25, Q75]	PEM-T median [Q25, Q75]
1	7.098e-01 [5.755e-01, 1.250e+00]	9.146e-01 [7.384e-01, 1.464e+00]
2	1.112e+05 [5.615e+04, 1.477e+05]	1.797e+05 [1.458e+05, 2.010e+05]
3	1.149e+02 [1.017e+02, 1.232e+02]	1.210e+02 [1.141e+02, 1.345e+02]
4	2.462e+02 [2.343e+02, 3.088e+02]	2.462e+02 [2.343e+02, 3.267e+02]
5	2.038e+01 [1.222e+01, 2.565e+01]	2.156e+01 [1.495e+01, 3.202e+01]
6	7.880e+01 [5.704e+01, 1.003e+02]	1.227e+02 [6.997e+01, 1.548e+02]
7	2.311e+01 [1.294e+01, 2.999e+01]	3.419e+01 [2.598e+01, 3.857e+01]
8	2.655e+02 [2.251e+02, 5.498e+02]	6.560e+02 [4.140e+02, 9.133e+02]
9	6.746e+02 [5.013e+02, 9.464e+02]	1.501e+03 [8.256e+02, 1.912e+03]
10	9.126e+04 [5.554e+04, 1.088e+05]	1.358e+05 [6.925e+04, 1.694e+05]
11	2.879e+02 [1.898e+02, 3.564e+02]	4.030e+02 [2.734e+02, 5.538e+02]
12	2.325e+07 [1.420e+07, 3.436e+07]	3.963e+07 [3.111e+07, 5.058e+07]
13	2.351e+02 [2.226e+02, 2.452e+02]	2.968e+02 [2.440e+02, 3.226e+02]
14	4.106e+00 [3.111e+00, 5.044e+00]	5.445e+00 [5.008e+00, 7.365e+00]
15	1.003e+02 [9.082e+01, 1.098e+02]	1.468e+02 [1.052e+02, 1.544e+02]
16	2.755e+01 [2.067e+01, 2.881e+01]	2.491e+01 [2.067e+01, 2.758e+01]
17	6.351e+00 [3.317e+00, 7.877e+00]	7.694e+00 [6.882e+00, 8.380e+00]
18	2.479e+01 [1.668e+01, 2.841e+01]	3.277e+01 [2.289e+01, 3.755e+01]
19	6.382e+00 [6.016e+00, 7.571e+00]	8.043e+00 [7.297e+00, 9.105e+00]
20	3.454e+00 [3.395e+00, 3.622e+00]	4.693e+00 [4.126e+00, 5.385e+00]
21	1.573e+01 [7.002e+00, 2.255e+01]	4.058e+01 [2.591e+01, 5.281e+01]
22	2.384e+01 [1.691e+01, 2.610e+01]	5.497e+01 [3.971e+01, 5.729e+01]
23	3.310e+00 [2.947e+00, 3.583e+00]	3.319e+00 [3.282e+00, 3.977e+00]
24	1.034e+02 [9.785e+01, 1.089e+02]	1.021e+02 [9.125e+01, 1.105e+02]

S4.20 Raw paid-run error dispersion, D=20

Absolute final error above the known optimum; all nine runs per function and method. This raw dispersion uses a different aggregation from the primary contrasts. Complete six-method IQR values are retained in internal_raw_error_iqr.csv. Scientific notation avoids rounding nonzero errors to zero.

Function	Fixed .1 median [Q25, Q75]	PEM-T median [Q25, Q75]
1	2.537e+00 [1.906e+00, 3.956e+00]	6.305e+00 [4.882e+00, 7.206e+00]
2	2.572e+05 [1.961e+05, 3.275e+05]	4.485e+05 [2.546e+05, 5.864e+05]
3	2.699e+02 [2.426e+02, 2.893e+02]	2.470e+02 [2.370e+02, 3.145e+02]
4	5.702e+02 [4.867e+02, 6.545e+02]	6.376e+02 [5.447e+02, 7.997e+02]
5	4.099e+01 [3.097e+01, 4.249e+01]	4.368e+01 [3.758e+01, 4.803e+01]
6	2.288e+02 [1.159e+02, 3.182e+02]	3.973e+02 [2.188e+02, 6.247e+02]
7	6.424e+01 [5.375e+01, 7.998e+01]	9.765e+01 [7.775e+01, 1.027e+02]
8	2.742e+03 [2.085e+03, 3.945e+03]	2.325e+03 [1.569e+03, 3.050e+03]
9	5.078e+03 [3.099e+03, 6.643e+03]	6.938e+03 [5.268e+03, 9.142e+03]
10	2.249e+05 [1.786e+05, 3.121e+05]	3.586e+05 [2.308e+05, 5.034e+05]
11	2.460e+02 [1.940e+02, 2.779e+02]	3.399e+02 [3.020e+02, 5.969e+02]
12	1.096e+08 [8.093e+07, 1.257e+08]	1.096e+08 [9.474e+07, 1.170e+08]
13	4.866e+02 [3.865e+02, 5.354e+02]	4.477e+02 [3.401e+02, 5.496e+02]
14	1.110e+01 [8.602e+00, 1.176e+01]	8.549e+00 [6.434e+00, 1.188e+01]
15	2.336e+02 [2.304e+02, 2.792e+02]	2.454e+02 [2.243e+02, 2.648e+02]
16	3.319e+01 [2.949e+01, 3.672e+01]	3.319e+01 [2.949e+01, 3.506e+01]
17	5.684e+00 [4.583e+00, 6.415e+00]	7.839e+00 [6.766e+00, 8.267e+00]
18	2.276e+01 [2.112e+01, 2.581e+01]	2.687e+01 [2.586e+01, 3.085e+01]
19	8.017e+00 [7.778e+00, 8.212e+00]	8.999e+00 [8.736e+00, 9.557e+00]
20	9.235e+01 [3.511e+01, 6.165e+02]	3.906e+02 [2.724e+02, 5.468e+02]
21	2.152e+01 [9.198e+00, 2.518e+01]	3.644e+01 [1.794e+01, 4.776e+01]
22	2.497e+01 [1.643e+01, 4.765e+01]	3.554e+01 [2.832e+01, 5.830e+01]
23	4.972e+00 [4.189e+00, 5.302e+00]	4.979e+00 [4.725e+00, 5.059e+00]
24	2.550e+02 [2.465e+02, 2.654e+02]	2.732e+02 [2.308e+02, 2.785e+02]

S4.40 Raw paid-run error dispersion, D=40

Absolute final error above the known optimum; all nine runs per function and method. This raw dispersion uses a different aggregation from the primary contrasts. Complete six-method IQR values are retained in internal_raw_error_iqr.csv. Scientific notation avoids rounding nonzero errors to zero.

Function	Fixed .1 median [Q25, Q75]	PEM-T median [Q25, Q75]
1	1.888e+01 [1.427e+01, 2.437e+01]	2.221e+01 [1.893e+01, 2.864e+01]
2	7.676e+05 [5.266e+05, 9.763e+05]	1.474e+06 [6.705e+05, 1.600e+06]
3	5.262e+02 [5.158e+02, 5.512e+02]	6.134e+02 [5.424e+02, 6.284e+02]
4	1.325e+03 [1.255e+03, 1.468e+03]	1.449e+03 [1.407e+03, 1.506e+03]
5	1.033e+02 [8.277e+01, 1.321e+02]	1.013e+02 [9.825e+01, 1.325e+02]
6	1.039e+03 [6.585e+02, 1.642e+03]	1.056e+03 [5.835e+02, 7.238e+03]
7	2.433e+02 [2.212e+02, 2.664e+02]	2.178e+02 [1.978e+02, 2.818e+02]
8	8.581e+03 [7.757e+03, 9.110e+03]	1.202e+04 [1.021e+04, 1.397e+04]
9	1.892e+04 [1.409e+04, 2.013e+04]	2.183e+04 [1.771e+04, 2.871e+04]
10	6.175e+05 [5.401e+05, 6.291e+05]	5.935e+05 [5.230e+05, 7.774e+05]
11	5.342e+02 [5.074e+02, 6.715e+02]	6.025e+02 [5.541e+02, 1.022e+03]
12	2.898e+08 [2.267e+08, 3.284e+08]	3.198e+08 [3.011e+08, 3.498e+08]
13	9.915e+02 [9.375e+02, 1.068e+03]	1.050e+03 [9.868e+02, 1.085e+03]
14	1.807e+01 [1.554e+01, 1.842e+01]	1.853e+01 [1.571e+01, 2.293e+01]
15	6.570e+02 [6.081e+02, 6.905e+02]	6.686e+02 [6.467e+02, 9.689e+02]
16	4.407e+01 [4.122e+01, 4.538e+01]	4.469e+01 [4.246e+01, 4.539e+01]
17	7.482e+00 [7.120e+00, 8.101e+00]	8.704e+00 [7.807e+00, 9.926e+00]
18	2.993e+01 [2.376e+01, 3.394e+01]	2.803e+01 [2.590e+01, 3.284e+01]
19	1.019e+01 [9.952e+00, 1.032e+01]	1.015e+01 [1.002e+01, 1.071e+01]
20	2.523e+03 [1.174e+03, 4.019e+03]	1.884e+03 [1.146e+03, 3.832e+03]
21	3.918e+01 [2.767e+01, 5.223e+01]	3.481e+01 [2.919e+01, 5.203e+01]
22	3.763e+01 [3.084e+01, 5.112e+01]	5.212e+01 [4.466e+01, 5.703e+01]
23	6.131e+00 [5.709e+00, 6.328e+00]	5.980e+00 [4.970e+00, 6.328e+00]
24	6.359e+02 [6.047e+02, 6.587e+02]	6.829e+02 [6.398e+02, 7.275e+02]

S5 Fixed BBOB structural groups

S1=f1-5 (separable), S2=f6-9 (low/moderate conditioning), S3=f10-14 (ill-conditioned unimodal), S4=f15-19 (structured multimodal), S5=f20-24 (weakly structured multimodal). Values are equal-function means of endpoint scores. These groups are descriptive and do not become new significant subgroup claims.

D	Group	Greedy	Fixed .1	Fixed .3	Random	PEM-T	PEM-ST
10	S1	-0.7844	-0.7450	-0.6725	-0.0863	-0.6413	-0.6387
10	S2	-1.8355	-1.8326	-1.7300	-0.2636	-1.5626	-1.6490
10	S3	-0.7892	-0.7628	-0.5927	-0.2560	-0.5328	-0.5780
10	S4	-0.3336	-0.2970	-0.2577	-0.1259	-0.2222	-0.2241
10	S5	-1.0173	-0.9644	-0.8509	-0.1286	-0.7885	-0.8232
20	S1	-0.8709	-0.8257	-0.7530	-0.1946	-0.7332	-0.7661
20	S2	-1.8155	-1.8273	-1.6113	-0.3130	-1.7193	-1.7759
20	S3	-0.8261	-0.7938	-0.6608	-0.3656	-0.6825	-0.6829
20	S4	-0.3937	-0.3728	-0.3356	-0.1352	-0.3228	-0.3479
20	S5	-0.7836	-0.7511	-0.6625	-0.1212	-0.6122	-0.5632
40	S1	-0.8011	-0.7802	-0.6967	-0.2467	-0.7166	-0.6799
40	S2	-1.6514	-1.6845	-1.5366	-0.4042	-1.5056	-1.4631
40	S3	-0.7668	-0.7300	-0.6667	-0.3791	-0.6789	-0.6440
40	S4	-0.3955	-0.3823	-0.3620	-0.1788	-0.3470	-0.3622
40	S5	-0.6313	-0.5684	-0.4519	-0.1309	-0.5548	-0.5230

S6 Localized bounds and metric conventions

Actual selected points; differences are ST minus T, within each dimension/history. Lower quantile loss does not establish optimization gains. State-level fallback and component-level aggregation differ; this table uses the frozen function-level diagnostic aggregation.

D	History	Loss ST-T	Margin ST-T	ESS ST-T	Fallback ST-T
10	fixed03	-0.0175	-0.1673	-5.5875	0.0000
10	greedy	-0.0106	-0.0884	-7.3423	0.0000
10	pool_random	-0.0033	-0.0317	-1.6307	0.0000
20	fixed03	-0.0250	-0.2231	-11.3516	0.0000
20	greedy	-0.0108	-0.1154	-13.5952	0.0000
20	pool_random	-0.0030	-0.0380	-1.4721	0.0000
40	fixed03	-0.0159	-0.1990	-11.1133	0.0000
40	greedy	-0.0082	-0.1211	-14.6521	0.0000
40	pool_random	-0.0020	-0.0260	-0.9110	0.0000

The frozen AUC and convergence curve floor absolute error at 1e-12 before division by initial error, then take log10, inclusive from n0 through B. The endpoint floors the error ratio after division. They are distinct formulas at the near-zero boundary. This dataset has no curve entries below -12 and no exactly optimal final run; the boundary distinction remains documented.

Full CSV sources include function_scores.csv, prespecified_function_contrasts.csv, internal_raw_error_iqr.csv, structural_endpoint_scores.csv, and localization_description.csv. Full checkpoint records and sparse declaration denominators accompany the audit. This printable supplement does not replace the machine-readable records or source fingerprint manifest.